

Circular Economy in Agriculture

Societal Internship Project (SIP)

Submitted to the Woxsen University, Hyderabad



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Date of Completion of the Project:

April, 2024

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Declaration

We hereby declare that the ELP Project entitled “ ”submitted to The School of Technology, Woxsen University, Hyderabad, Telangana State, India, in partial fulfilment of the requirements for the project is the original and independent work carried out by us, under the supervision of **Dr. Riya Bhattacharya**, Assistant Professor of Chemistry in the School of Technology, Woxsen University, Hyderabad. This Project has not formed the basis for the award of any Projects/Degree /Diploma /Fellowship /Associateship of this University or any other institution.

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Place: Hyderabad

Date: 30.04.2024

Certificate

This is to certify that the work embodied in this Project entitled "Personal Protective Equipment in Industrial Premises" has been carried out by **Sarva Advaith Narayana, Satya Sankeerth Thalikota, Shashank Pullabhatla Polamada Aswattha Vagdhesh Vikram** under my diligent supervision at the School of Technology, Woxsen University, Hyderabad, Telangana State, India.

The project was conducted in the complete fulfilment of the requirements for the **Experiential Learning Project (ELP)**, and the findings presented in this document are original and have not been submitted, in whole or in part, for any projects/degree at any other University or educational institution.

I attest that the students named above have conducted the research and analysis diligently and have adhered to the academic and research community's ethical standards and guidelines.

Furthermore, I affirm that this project work has not been previously submitted for assessment, and the intellectual contributions of the students are genuine and reflective of their efforts.

Place: Hyderabad

Date: 30.04.2023

Dr. Riya Bhattacharya

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Acknowledgement

We extend our deepest gratitude to, Dr. Riya Bhattacharya, Assistant Professor at the School of Technology, Hyderabad, Telangana State, India, who served as our project supervisor. He played a pivotal role in guiding us from the inception of the research topic to providing continuous support and valuable advice throughout every stage of the project. His constructive suggestions, unwavering inspiration, encouragement, and engaging discussions, including brainstorming sessions, were instrumental in our successful and efficient completion of the project.

We would also like to express our appreciation to all the faculty members of the School of Technology, especially **Dr. Kiran Kumar R**, Dean, and **Dr. Shaik Ahmed**, Programme Director, for their unwavering support whenever we sought assistance.

A special acknowledgement goes to **Local Farmers**, in the Medak district for their cooperation, fostering beautiful communication, engaging in valuable discussions, and contributing to an experiential atmosphere through the visit.

Our sincere thanks are extended to Woxsen University, Hyderabad, for their support in facilitating the smooth conduct of our project work at the School of Technology.

Finally, we acknowledge and express gratitude to the Almighty for the blessings that have made our dreams come to fruition.

4. Abstract

Examine Current Agricultural Waste Practices: Review the existing methods of handling agricultural waste to gain insights into their environmental implications.

Determine Weaknesses and Challenges: Identify where current waste management systems fall short, focusing on areas with negative environmental, economic, or social effects.

Suggest Innovative and Sustainable Solutions: Create a strategic plan to minimize waste and maximize resource recovery, emphasizing a circular economy approach in agriculture.

Educate Farmers on Circular Economy Practices: Develop training and educational programs to help farmers implement circular economy principles in their operations.

Encourage Collaboration and Partnerships: Establish partnerships among farmers, researchers, and businesses to facilitate knowledge sharing and promote sustainable waste management.

Keywords :

Circular Economy (CE)

Linear Economy

Agriculture

Sustainability

Resource Efficiency

Waste Reduction

Sustainable Development Goals (SDGs)

Key Performance Indicators (KPIs)

5. Introduction

Agriculture plays a fundamental role in society, yet traditional farming practices often lead to waste and unsustainable resource use. The Circular Economy (CE) is an approach that seeks to break this cycle by promoting recycling, waste reduction, and the creation of value from byproducts. This project aims to investigate how the circular economy can be implemented in agriculture, demonstrating the benefits of this approach and discussing the potential challenges.

The document will explore how agricultural practices can be restructured to align with circular economy principles, offering a detailed strategy that encompasses every stage of farming, from soil preparation to harvesting and beyond. Through the application of CE methods, we can decrease environmental impact while creating a more robust and efficient agricultural system.

Additionally, the project will propose a five-year plan to guide the implementation of circular economy practices in agriculture, complete with specific objectives and Key Performance Indicators (KPIs) to measure success. The alignment with the United Nations Sustainable Development Goals (SDGs) will also be discussed, showing how the circular economy in agriculture can contribute to broader global sustainability efforts.

6. Field Visit and Learnings:

As part of the Societal Internship Project (SIP), we have visited local farms of Medak district, Telangana and through these we recorded some intuitive observations from seeing the crops and farm conditions to interacting with



Fig 0.0 Farmer

farmers and conducting surveys and finding out their problems through the field and their methods of farming they used to cultivate the crop. We have asked them several questions related to waste management systems, crop rotation, Livestock management systems and much more. Our main observation was that nowadays farmers tend to know good farming methods but do not follow these methods. Most of them conveyed that new-aged farmers are lazy and stubborn and may also see a sign of ease for easy growing crops to make money rather

than working hard on it. The cost of labour, machinery, seeds, fertilizers, pesticides, etc, has increased and not getting good prices for farmers was one of their concerns discussed with us. Another main concerns we have discussed about incineration of stubble crops as many had harvested rice crops and incinerated them. Some also say that burning them increases fertility, some say that the pests inside will only die if burnt. We are yet to discover the truth on these topics. Here are some of the photos taken during our visit.



Fig 0.1 New Technique of Rice Crop farming



Fig 0.2 Burnt Rice Crop



Fig 0.3 Unburnt Rice Crop

4. Feedback Form

స.నెం	ప్రశ్నలు	రైతుల పేర్లు		
1.	మీ పేరు ఏమిటి ?	రాములు	పెద్దులు	పోషయ్య
2.	ఈసారి ఏ పంట వేశారు?			
3.	మీకు పశువులు ఉన్నాయా			
4.	మీరు మీ పంటను ఎలా పండిస్తారు (యంత్రంతో లేదా యంత్రం లేకుండా)			
5.	మీ పొలానికి ఇక్కడ నీరు, కూలీలు అందుబాటులో ఉన్నాయా?			
6.	మీరు మీ పంటను దహనం చేస్తారా?			
7.	మీ పొలం నుండి మీకు ఏ వ్యర్థాలు వస్తాయి మరియు మీరు వ్యర్థాలను ఏమి చేస్తారు?			
8.	మీరు పంట మార్పిడిని అనుసరిస్తున్నారా?			
9.	మీరు పరిపూరకరమైన వ్యవసాయ పద్ధతులను అర్థం చేసుకున్నారా?			

స.నెం	ప్రశ్నలు			
9.	మీరు పరిపూరకరమైన వ్యవసాయ పద్ధతులను అనుసరిస్తున్నారా? లేకపోతే, ఎందుకు?			
10.	మీరు ఏదైనా ఎరువులు వాడుతున్నారా, అది సేంద్రీయమైనదా లేదా?			
11.	మీరు పెస్ట్ కంట్రోల్ సిస్టమ్స్ ఉపయోగిస్తున్నారా?			
12.	మీకు ఏ అగ్రోటెక్నాలజీ తెలుసా?			
13.	మీరు ఎంత దిగుబడిని పొందుతారు (సగటున.) మరియు అది మీ పొలానికి సరిపోతుంది?			
14.	మీరు ఏ స్థలంలో మీకు వస్తువులను మరియు ఎవరికి విక్రయిస్తారు?			
15.	మీరు చివరిసారిగా అమ్మిన పంట ధర ఎంత (పొలం ధరతో పోల్చండి)?			
16.	మీకు ప్రభుత్వ పథకాలు అర్థం అవుతున్నాయా?			

7. Literature Review:

In this project, we have reviewed several papers based on Agricultural waste management systems, Linear economy, Circular economy and new aged systems and certain governments, collaborative government councils like the European Union, and UN organisations implementing these Circular economies. We have also cited the UN's official website stating and comparing SDGs and Circular Economy. The Circular economy is a hot topic among researchers and there are thousands of research based on CE practices, challenges and Implementation of CE principles in agriculture. Supranational bodies have also incorporated circular economy concerns, most notably the 2015 EU Circular Economy Strategy and many more countries like China.

Papers	Authors	Outcomes
Agricultural waste management strategies for environmental sustainability	(Koul et al., 2022)	Various products like Biochar and biofuel generation, etc.. Different types of agro waste.
Agricultural waste: Review of the evolution, approaches and perspectives on alternative uses	(Duque-Acevedo et al., 2020)	Gives an analysis of agricultural waste and the steps required to research
Circular economy in agriculture. An analysis of the state of research based on the life cycle	(Velasco-Muñoz et al., 2022)	It gives analysis about Circular economy
Long-Term Evidence Shows that Crop-Rotation Diversification Increases Agricultural Resilience to Adverse Growing Conditions in North America	(Bowles et al., 2020)	Crop rotation and its benefits
Agricultural Waste in Circular Economy: An Indian Scenario	(Baruah et al., 2024)	India and its implementation of CE

8. Linear Economy and Its drawbacks:

Linear Economy in agriculture is a new-aged traditional method and approach towards farming, it works on the principle of the “take-make-and-dispose” model. It requires a lot of resources to run these models leading to the generation of waste through different stages/processes in agriculture. It leads

to environmental damage, and soil degradation and is not eco-friendly making it not a good and sustainable process for the future increase of Food consumption and less waste generation (Sariatli, 2017).

Characteristics in Linear Economy:

1. Resource extraction: We need raw materials from outside such as fertilizers, pesticides, etc.
2. High Input use: More synthetic inputs and inorganic inputs are given to the soil for different uses like Pests, and unwanted herbs.
3. Single-use resources: The used resources are thrown away, they either get filled up landfills or if not incinerated release a lot of pollution and dust into the air and leave unwanted ash too. This might include crop residues, packaging, and other agricultural byproducts.
4. Waste generation is very high in a Linear Economical way

Environmental Impacts through the following Linear Economy: This leads to soil degradation, polluting land, water and soil. It Leads to contamination of resources and this also increases Carbon emission into the atmosphere. Additionally, the disposal of agricultural waste contributes to pollution

Economic Challenges: This increases the annual budget for the farmers through expensive inorganic yet unwanted materials being injected into the soil. The high resource use and waste generation in a linear system also raise sustainability concerns, as these practices are not conducive to long-term productivity and resilience.

9. Circular Economy and Its Principles:

A circular economy (CE) is a sustainability-focused approach that seeks to reduce waste by reusing and recycling resources, creating a closed-loop system. Unlike the traditional linear economy's "take-make-dispose" model, CE aims to keep materials in use for as long as possible. In agriculture, this means turning waste into valuable resources like compost or bioenergy, minimizing the need for new inputs, and promoting practices that regenerate the soil. This approach benefits both the environment and the economy by

reducing pollution, lowering costs, and supporting resource efficiency.(Velasco-Muñoz et al., 2021).

3 Principles of CE in agriculture:

Firstly, 1. Efficient Use of resources, 2. Optimization of process, 3. Prevent wastage Secondly, 1. Environmental, 2. Economic, 1. Social sustainability Finally, To Create a regenerative system that enables the closure of nutrient loops and minimizes leakages.

Implementation of CE principles in Agriculture:

1. Production and Planning: Should know the knowledge of the demand for food and resources.
2. Production organization: Correct way of growing crops and no negative impacts on the crops to increase yield and less wastage.
3. Production application: Planting, harvesting, and evaluation of possible errors throughout the process.

10. Circular Economy over Linear Economy:

We will now discuss about benefits of CE over LE to increase yield and decrease waste. LE depletes resources and generates waste, while CE focuses on regeneration, recycling, and sustainability (Sariatli, 2017).

Aspects	Linear Economy (LE)	Circular Economy (CE)
Resource Use	Extract resources, use them, then discard them	Reuse, recycle, and regenerate resources
Waste Management	Waste is typically sent to landfills or incinerated	Waste is typically sent to landfills or incinerated
Environmental Impact	High due to reliance on synthetic inputs and waste disposal	Reduced impact by focusing on sustainable practices

Soil Health	Tends to degrade soil quality with heavy chemical use	Prioritizes soil regeneration and health through composting and cover crops
Energy Use	May or May not Rely on non-renewable energy sources	Encourages renewable energy, such as bioenergy
Sustainability	Generally unsustainable due to resource depletion	Aims to create a self-sustaining system by recycling and reducing waste
Economic Approach	Focuses on short-term gains with heavy resource use	Seeks long-term sustainability and cost savings by reusing resources

11. Circular Economy and its challenges and overcoming it:

A good and futuristic model too has some disadvantages and challenges, and we will discuss them. We will also discuss the strategies to minimize those challenges.

The Circular Economy (CE) model faces several hurdles. Regulatory constraints often limit how much companies can embrace circular practices. Optimizing reverse logistics—where materials are returned for recycling or reuse—can be complex and costly. The wide geographic spread of businesses complicates efficient material collection and redistribution. Additionally, defining system boundaries is tricky, leading to potential leakage where materials escape the circular loop. These challenges require innovative solutions and collaborative efforts to realize the benefits of a circular economy fully.

To overcome the challenges in implementing the Circular Economy (CE), consider these strategies:

Maximizing Resource Utilization: By narrowing resource loops, you ensure that every resource is used to its fullest potential. This involves optimizing agricultural practices to reduce waste and make the most of existing resources.

Extending Product Lifespan: Slowing resource loops means extending the life of agricultural products and equipment. This strategy emphasizes maintenance, repair, and repurposing, allowing farmers to use their resources longer and reduce the need for new inputs.

Recycling and Reusing Materials: Closing resource loops involves creating systems where agricultural materials are recycled and reused, minimizing waste. This can include composting organic waste, reusing equipment, and repurposing byproducts.

Regenerating Resource Flows: Regeneration focuses on restoring resources to their natural state. This strategy emphasizes practices like composting and cover cropping, which help regenerate soil health and support sustainable farming.

12. Methodology

To thoroughly evaluate Circular Economy (CE) practices in agriculture, we used both qualitative and quantitative methods. The qualitative approach centered on field visits, interviews, and surveys with farmers in Medak district, Telangana, providing insights into their agricultural techniques, challenges, and views on sustainable practices. We sought to understand farmers' perspectives on waste management, crop rotation, and livestock practices, while also exploring the socio-economic factors that influence their decisions.

On the quantitative side, we gathered and analyzed data on farming operations, including resource usage, crop yields, waste production, and energy consumption. This data collection helped us establish a baseline for

comparison and assess the effectiveness of CE strategies over time. By blending qualitative observations with quantitative data, our goal was to create a well-rounded picture of the difficulties and prospects in adopting Circular Economy approaches in agriculture.

12. Sustainable Development Goals

SDG 12: Responsible Consumption and Production

The Circular Economy in agriculture aligns with SDG 12 by promoting recycling, reducing waste, and encouraging sustainable resource use for production processes.

SDG 13: Climate Action

Circular agricultural practices like composting, no-till farming, and bioenergy reduce greenhouse gas emissions, helping to combat climate change in line with SDG 13.

SDG 15: Life on Land

Supporting SDG 15, the Circular Economy in agriculture fosters biodiversity, reduces soil degradation, and promotes healthier ecosystems through practices like crop rotation and integrated pest management (UN/DESA Policy Brief #105: Circular Agriculture for Sustainable Rural Development | Department of Economic and Social Affairs, n.d.).

13. Stages in the Agro-food Industry and markets and their CE Implementation:

Field preparation: Reuse spent growing media, recycle greenhouse components, and extend the lifespan of soilless crop substrates. Additionally, focusing on reducing waste and promoting sustainability.

Sowing: Focus on reusing nursery materials, and developing more sustainable crops through varietal improvement. Integrated value chains should be established to encourage systemic circular practices.

Fertilizer Preparation: Focus on optimizing fertilizer efficiency, using organic fertilizers, and standardizing nutrient circularity indicators to minimize environmental impact. Integrate agronomic research with a circular economy perspective for sustainable nutrient management.

Pest management: Emphasize integrated pest control, biological methods, and natural solutions over chemical products.

Mulching: Prioritize organic materials from the farm or recycled materials from other sources. If that's not feasible, use biodegradable mulch films made from renewable resources. This approach reduces waste and promotes sustainability by eliminating non-renewable inputs.

Circular Economy: Using efficient systems like drip and sprinkler, which significantly reduce water use. Incorporating hydroponics and aquaponics allows water recirculation and resource reuse. Additionally, using wastewater for irrigation provides nutrients and supports traditional ferti-irrigation methods.

Pruning and Training: Focus on replacing plastic materials with biodegradable alternatives to guide plant growth. Recycling plastic materials for use in other sectors can reduce waste while composting and bioenergy production from pruning waste supports sustainability.

Harvesting: Focus on reusing tools, machinery, and packaging materials. You can collaboratively as a whole village to use the same type of machinery and tools, if possible, to decrease carbon emissions.

Waste management:, Prioritize on-farm reuse to minimize waste transport and close nutrient loops. Develop interconnected networks for resource sharing across sectors, reducing environmental impact and fostering new business opportunities. CE promotes sustainable practices, job creation, and improved resource efficiency.

14. Key Performance Indicators (KPI)

Key Performance Indicators (KPIs) for implementing Circular Economy (CE) in agriculture should focus on resource efficiency, waste reduction, recycling, and overall sustainability. Based on the summarized thread, here are some new KPIs along with their calculation methods:

KPI 1: Waste Reduction

- **Description:** Measures the reduction in agricultural waste generated over time.
- **Calculation:** Compare the total weight or volume of waste before and after implementing CE practices. Express this as a percentage reduction.

KPI 2: Resource Recovery Rate

- **Description:** Assesses the proportion of waste that is recycled, reused, or composted.
- **Calculation:** Divide the total amount of waste that is recycled, reused, or composted by the total waste generated. Express this as a percentage.

KPI 3: Fertilizer Efficiency

- **Description:** Measures the efficiency of fertilizer use.
- **Calculation:** Compare the amount of fertilizer used with crop yield. Calculate the ratio of fertilizer input (kg) to crop output (tons) to determine efficiency.

KPI 4: Water Usage Efficiency

- **Description:** Evaluate the efficiency of water use in irrigation systems.
- **Calculation:** Measure the amount of water used per hectare and compare it to crop yield. A lower ratio indicates greater efficiency.

KPI 5: Energy Usage Reduction

- **Description:** Assesses the reduction in energy use through CE practices.
- **Calculation:** Track energy consumption before and after implementing CE measures. Calculate the percentage reduction in energy use.

KPI 6: Bioenergy Production

- **Description:** Measures the amount of energy generated from agricultural waste.
- **Calculation:** Determine the total energy output from bioenergy sources like biogas. Express this in kilowatt-hours (kWh) or megajoules (MJ).

KPI 7: Soil Health Improvement

- **Description:** Assesses changes in soil health and fertility.
- **Calculation:** Use soil tests to measure organic matter content, pH, and nutrient levels. Track changes over time to evaluate improvement.

KPI 8: Greenhouse Gas Emissions Reduction

- **Description:** Evaluate the reduction in greenhouse gas emissions.
- **Calculation:** Measure emissions before and after CE implementation. Express the reduction as a percentage decrease in carbon dioxide equivalents (CO₂e).

KPI 9: Farm Revenue from Circular Practices

- **Description:** Measures the revenue generated from circular economy activities.
- **Calculation:** Calculate the income from activities like selling compost, bioenergy, or recycled products. Express this as a percentage of total farm revenue.

15. Complementary Crops in Crop Rotation:

Complementary crops in crop rotation involve alternating different types of crops in a strategic sequence to boost soil health, manage nutrients, and curb pests and diseases. This approach combines the strengths of traditional crop rotation with the benefits of companion planting, fostering a more sustainable agricultural system.

Crop rotation involves growing different crops on the same field over several seasons to prevent soil exhaustion and disrupt pest cycles. Complementary

crops add an extra layer by considering how various plants can work together to improve each other's growth. Benefits include improved soil fertility, natural pest control, reduced weed growth, and better resource use.

1. Corn and Soybeans

Corn Waste: Corn stalks and cobs can be used as mulch or composted to add organic matter to the soil.

Benefit for Soybeans: The organic matter from corn waste improves soil structure and fertility, supporting soybean growth.

2. Wheat and Barley

Wheat Waste: Wheat straw can be used as animal bedding or mulch, and can be composted to enrich the soil.

Benefit for Barley: The composted wheat straw adds nutrients to the soil, benefiting barley, which also has similar nutrient requirements.

3. Beans and Corn

Bean Waste: Bean plants fix nitrogen in the soil, and their waste can be composted or used as green manure.

Benefit for Corn: The added nitrogen from bean waste helps improve soil fertility for corn, which is a heavy feeder.

4. Rice and Vegetables

Rice Waste: Rice straw can be composted or used as mulch to improve soil structure and moisture retention.

Benefit for Vegetables: The organic matter from rice waste benefits leafy greens, tomatoes, and other vegetables, providing a sustainable source of nutrients.

5. Potatoes and Carrots

Potato Waste: Potato plant tops and other organic waste can be composted or used as green manure.

Benefit for Carrots: The composted potato waste adds organic matter to the soil, supporting the growth of carrots, which benefit from well-structured soil.

6. Tomatoes and Peppers

Tomato Waste: Tomato plants produce a lot of organic waste, which can be composted and used to enrich the soil.

Benefit for Peppers: The composted tomato waste provides nutrients and organic matter to support pepper growth.

7. Lettuce and Cabbage

Lettuce Waste: Lettuce leaves and stems can be composted or used as green manure.

Benefit for Cabbage: The organic matter from composted lettuce waste benefits cabbage by improving soil structure and nutrient content.

8. Squash and Beans

Squash Waste: Squash vines and other organic waste can be composted or used as mulch.

Benefit for Beans: The composted squash waste provides nutrients and organic matter, supporting bean growth and soil health.

9. Cucumbers and Spinach

Cucumber Waste: Cucumber vines and leaves can be composted to create nutrient-rich compost.

Benefit for Spinach: The composted cucumber waste adds organic matter to the soil, enhancing spinach growth.

17. Integrated CE Business Model :

Year 1: Start with Diverse Crop Rotation and Focus on Soil Health

Complementary Crops and Crop Rotation: Begin with wheat and legumes (e.g., chickpeas) for the first cycle, followed by maize and sunflowers for the second cycle. This rotation helps maintain soil health and provides diversity.

Off-Season Crops: Plant cover crops such as clover or ryegrass to prevent soil erosion and add organic matter during the off-season.

KPIs to Track: Fertilizer Efficiency, Soil Health Improvement, Waste Reduction, Resource Recovery Rate.

CE Strategies: Use composting to enhance soil health, rotate complementary crops, and utilize cover crops during off-season to reduce nutrient loss and control weeds.

Year 2: Expand Crop Diversity and Implement Sustainable Irrigation

Complementary Crops and Crop Rotation: Plant rice and lentils for the first cycle, followed by wheat and soybeans for the second cycle. This combination promotes nitrogen fixation and reduces the need for synthetic fertilizers.

Off-Season Crops: Use green manure crops like vetch or alfalfa to add nutrients to the soil during the off-season.

KPIs to Track: Water Usage Efficiency, Energy Usage Reduction, Greenhouse Gas Emissions Reduction.

CE Strategies: Optimize irrigation with drip systems to reduce water use, promote energy efficiency through renewable sources, and use green manure to maintain soil health.

Year 3: Enhance Resource Recovery and Focus on Bioenergy Production

Complementary Crops and Crop Rotation: Rotate wheat and oats for the first cycle, followed by maize and sorghum for the second cycle. This rotation supports resource recovery and promotes soil structure.

Off-Season Crops: Plant legumes like lupine or field peas to increase nitrogen in the soil during the off-season.

KPIs to Track: Resource Recovery Rate, Bioenergy Production, Farm Revenue from Circular Practices.

CE Strategies: Convert crop residues into bioenergy, compost organic waste, and promote resource recovery by reusing agricultural byproducts.

Year 4: Foster Partnerships and Engage in Community Outreach

Complementary Crops and Crop Rotation: Plant barley and beans for the first cycle, followed by wheat and sunflower for the second cycle. This rotation increases biodiversity and provides a varied range of crops.

Off-Season Crops: Use mustard or buckwheat to suppress weeds and improve soil structure during the off-season.

KPIs to Track: Partnerships Established, Job Creation and Rural Development, Community Engagement.

CE Strategies: Establish partnerships with local businesses, engage in community outreach, and continue resource recovery practices to reduce waste.

Year 5: Advocate for Circular Economy Policies and Focus on Continuous Improvement

Complementary Crops and Crop Rotation: Rotate rice and groundnut for the first cycle, followed by wheat and canola for the second cycle. This promotes oilseed production and improves resource efficiency.

Off-Season Crops: Plant cover crops like rye or crimson clover to maintain soil health during the off-season.

KPIs to Track: Waste Reduction, Resource Recovery Rate, Partnerships Established, Farm Revenue from Circular Practices.

CE Strategies: Advocate for CE policies, focus on continuous improvement based on KPIs, and continue promoting off-season cover crops to enhance sustainability.

18. Expected Outcomes

The expected outcomes of this project include a more sustainable agricultural system where resources are reused and waste is minimized. Farmers will benefit from circular economy practices, leading to reduced costs for inputs

like fertilizers and pesticides. These practices also contribute to lower greenhouse gas emissions, improved soil health, and decreased reliance on non-renewable energy sources. The project aims to create a more resilient and efficient farming approach by promoting circular economy principles, benefiting both the environment and farm profitability.

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